Journal of Behavioral and Experimental Finance 44 (2024) 100991



Contents lists available at ScienceDirect

Journal of Behavioral and Experimental Finance

journal homepage: www.elsevier.com/locate/jbef





Google search and cross-section of cryptocurrency returns and trading activities

Lai Hoang [a], Duc Hong Vo [a,b,*]

[a] UWA Business School, The University of Western Australia, Australia
[b] Research Center in Business, Economics & Resources Ho Chi Minh City Open University, Viet Nam

ARTICLE INFO

JEL classification:
G12
G15
G17

Keywords:
Google search volume
cryptocurrency
price pressure hypothesis
investor attention

ABSTRACT

This paper examines the effect of investor attention on the cross-section of cryptocurrency returns and trading activities. We find that cryptocurrencies associated with higher abnormal Google search volume subsequently exhibit higher returns, higher volatility, and higher trading volume. The results are robust to alternative sample periods and alternative search keywords, providing concrete support to the attention-induced price pressure hypothesis and consistent with prior studies on the equity market. The effect is more pronounced among larger cryptocurrencies. Only a partial reversal after the initial return increase is observed, implying that investor attention permanently impacts cryptocurrency prices.

1. Introduction

Since the introduction of Bitcoin in 2009, cryptocurrencies have quickly grown into a significant alternative asset class. Based on the data from coinmarketcap.com, by the end of May 2024, there are more than 10,000 cryptocurrencies actively traded on more than 500 exchanges worldwide, with a total market capitalization of more than $2500 billion. Consequently, academic research on cryptocurrencies has also proliferated. While early studies mainly focus on Bitcoin and several other largest cryptocurrencies (Kristoufek, 2013; Urquhart, 2016; Yermack, 2015; Cheah and Fry, 2015), recent studies have started paying attention to the entire universe by applying standard asset pricing tools to analyse the cross-section of cryptocurrencies. They found that while some factors capturing the cross-section of cryptocurrency returns also perform similarly in the cryptocurrency market (e.g., size (Li et al., 2020, Liu et al., 2022), momentum (Liu et al., 2022), liquidity (Zhang and Li, 2023), and downside risk (Zhang et al., 2021)), some others do not (e.g., MAX effect (Li et al., 2021), beta, volatility, and price (Liu et al., 2022)). This paper extends this strand of research by examining whether investor attention predicts the cross-section of future returns and trading activities in the cryptocurrency market.

The efficient market hypothesis (EMH), introduced by Fama (1970), posits that when information arrives in the market, investors process it promptly and rationally, resulting in current prices accurately reflective of all available information. An important implicit assumption of the EMH is that investors pay sufficient attention to all incoming information. However, this might not always be the case, as psychological research suggests that attention is a scarce cognitive resource (Grossman and Stiglitz, 1980; Kahneman, 1973). In other words, investors have limited attention, which prevents them from processing information effectively. Retail investors are more susceptible to this scarcity than institutional investors due to their relatively limited resources (Barber and Odean, 2008). Scarcity is increasingly constrained in the internet era with a plethora of easily accessible information (Smales, 2022). This might lead to a less efficient market as information is only incorporated into prices when investors start paying attention to it (Huberman and Regev, 2001), indicating that limited attention is a determinant of asset price.

Barber and Odean (2008) develop a model that explains the relationship between retail investors' attention and the *direction* of the subsequent price movement. Specifically, retail investors have to search through numerous stocks to collect information supporting their investment decisions for stock purchases. In contrast, investors do not face the same problem when selling as they only possess a limited number of stocks for sale. The constraints facing retail investors do not apply to institutional investors for two reasons. First, institutional investors usually hold a large number of stocks in their portfolios. As such, they face a similar search problem when they sell and buy. In addition, the

* Corresponding author at: UWA Business School, The University of Western Australia, Australia
E-mail address: duc.vo@uwa.edu.au (D.H. Vo).

https://doi.org/10.1016/j.jbef.2024.100991
Received 13 November 2023; Received in revised form 20 August 2024; Accepted 14 September 2024
Available online 26 September 2024
2214-6350/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

ability to sell short further blurs the difference between the number of stocks to buy and to sell. Second, institutional investors devote more time and resources to search for stocks to buy and sell than do retail investors. For example, institutional investors usually have exclusive access to sophisticated information services such as Reuters or Bloomberg terminals. Consequently, when retail investors pay attention to a stock, they are likely to purchase it eventually, creating upward pressure on the price of a relevant stock. We refer to this mechanism as the "attention-induced price pressure hypothesis".[1] Barber and Odean (2008) also find empirical evidence supporting the model prediction using proprietary data that includes trading and position records for the investments of 78,000 households from January 1991 to December 1996. The authors show that individual investors tend to be net buyers of attention-grabbing stocks, where attention is indirectly proxied by high abnormal trading volume, extreme one day returns, and media appearances. Da et al. (2011) use Google search volume as a direct measure of retail investor attention and, consistent with Barber and Odean (2008), find that that investors buying attention-induced stocks enjoy a short-run increase in prices. However, stock prices experience an eventual reversal, indicating that the price pressure triggered by investor attention is temporary.

Following studies using the same method also find consistent results in various non-US stock markets, including Germany (Bank et al., 2011), France (Aouadi et al., 2013), Japan (Adachi et al., 2017; Takeda and Wakao, 2014), Vietnam (Nguyen et al., 2020), and Australia (Chai et al., 2021). Google search volume is one of the most common gauges of retail investors' attention due to three reasons. First, internet search is a *revealed* attention measure: people searching for something on the internet are undoubtedly paying attention to it. Second, with Google handling nearly 90 % of global internet searches,[2] It is likely representative of the general population's online search behaviour. Third, Google, as a "free" source, is more likely the tool used by retail investors to search for financial information, while institutional investors usually have access to more sophisticated information services.

Although the relationship between retail investor attention and future stock returns has been tested predominantly on stock markets, the cryptocurrency market provides a unique context to reexamine this relation for three reasons. First, unlike stock markets where institutional investors are majority owners,[3] cryptocurrencies are dominated by retail investors who rely much more on free search engines such as Google to collect information. Second, due to the lack of fundamentals (or at least it is extremely difficult to determine fundamentals), the cryptocurrency market is free of confounding factors that potentially drive both stock returns and investor attention, such as earnings announcements, insider transactions, or in general the release of any price-relevant information, rendering the relationship between investor attention and subsequent price movement more likely to be causal if it exists. Third, while finance researchers tend to agree towards weak or semi-strong forms of market efficiency in the stock market (Polyzos et al., 2024), the cryptocurrency market is usually considered much less inefficient. For example, Wei (2018) shows that many cryptocurrencies exhibit significant autocorrelation and non-independence. Other studies find significant profitability of various technical trading rules for Bitcoin (Gerritsen et al., 2020; Corbet et al., 2019) and other main cryptocurrencies (Grobys et al., 2020; Ahmed et al., 2020; Svogun and Bazán-Palomino, 2022); thus challenging even the weak form of the EMH. Meanwhile, some studies document time-varying and state-dependent market efficiency (Noda, 2021; López-Martín et al., 2021; Polyzos et al., 2024), which is consistent with the adaptive market hypothesis (AMH), put forth by Lo (2004).[4] Such inefficiency implies that cryptocurrency prices are more likely to be subject to behavioural biases such as limited attention,[5] which is arguably a consequence of the difficulty in estimating intrinsic value and the dominance of retail investors.

Using a sample of tradable cryptocurrencies from January 2018 to December 2022 in univariate portfolio analysis, we show that a portfolio of cryptocurrencies with higher abnormal Google search volume exhibits higher future returns in the subsequent week. The difference in returns of portfolios with the highest and lowest abnormal search volume is significantly positive. Furthermore, we control for factors suggested by the established theories to ensure that the effect of investor attention is most likely causal. Indeed, the effect of investor attention on cryptocurrency returns and trading activities may be spurious due to confounding factors. For example, as some cryptocurrencies may lack sufficient liquidity to absorb increased trading activity without substantial price movements, the impact of investor attention might be attributed to illiquidity, which can exacerbate price deviations. First, following the capital asset pricing model (CAPM), we control the systematic risk, as measured by beta, capturing the relationship between the asset's and the market's returns. As highlighted by Fama and French (1992), we also consider market capitalization because smaller-market-capitalization cryptocurrencies tend to have higher returns to compensate for their higher risk. In addition, the momentum is controlled because Carhart's (1997) four-factor model suggests that stocks that have performed well (or poorly) in the recent past will continue performing well (or poorly) in the near future. Next, we consider that the idiosyncratic volatility, representing the risk specific to an individual asset, is not explained by overall market movements. Higher volatility can lead to higher returns (and vice versa), which aligns with the risk-return trade-off. Then, extreme positive returns are considered, capturing investors' preference for assets with lottery-like payoffs. This concept is linked to behavioural finance theories, where investors are attracted to assets with the potential for high payoffs, impacting asset pricing and trading behaviour, as discussed by Bali et al. (2011). According to liquidity theory, less liquid assets require higher returns to compensate investors for higher transaction costs and the risk of being unable to trade quickly. Hence, Amihud (2002) illiquidity ratio is controlled. Finally, following Harvey and Siddique (2000), the co-skewness between cryptocurrency and market returns is controlled. Using the bivariate portfolio analysis and the Fama and Macbeth (1973) regression, we find that our results are robust after controlling for these confounding factors.

Although the main results are consistent with prior studies on stock markets, we find a contrary effect of size. Specifically, while the positive relation between Google search volume and subsequent returns remains in all size subsamples, it is more pronounced among large cryptocurrencies. This is inconsistent with Da et al. (2011) and Takeda and Wakao (2014), who show that the effect of Google search volume on stock returns is stronger among small firms. We attribute this difference to several unique characteristics of cryptocurrencies, including (i) the dominance of retail investors, (ii) the transparency of blockchain and the lack of fundamentals which deteriorate the information advantage of institutional investors, and (iii) the popularity of large cryptocurrencies,

[1] Although Barber and Odean's (2008) study does not specifically name the model, the term is frequently used by several following studies, for example, Da et al. (2011), Joseph et al. (2011), Takeda and Wakao (2014), and Yung and Nafar (2017).

[2] https://www.statista.com/statistics/216573/worldwide-market-share-of-search-engines/

[3] For example, institutional investors are estimated to own about 80 % of all stocks in the S&P500. See also https://hbr.org/2019/02/how-big-a-problem-is-it-that-a-few-shareholders-own-stock-in-so-many-competing-companies.

[4] In contrast to the EMH, the AMH suggests that investors are not always rational, and markets are not always efficient. Instead, investors are information-intensive. Thus, they absorb new market information and adjust their investment strategies to adapt to new market dynamics.

[5] Some other behavioural biases that affect cryptocurrency market activities include herding (Youssef and Waked, 2022) and investor sentiment (Naeem et al., 2021).

leading to a more solid link between the search queries and subsequent purchases.[6] We also find a reversal in the long run as Google search volume is negatively related to future returns over the next year. However, the reversal is only partial, unlike a full reversal observed in the U.S. stock market (Da et al., 2011).

As a natural extension of the analysis, we examine whether Google search volume affects future return volatility and trading volume. We find that cryptocurrencies with higher Google search volume are traded more and exhibit higher volatility in the subsequent week, further supporting the attention-induced price pressure hypothesis. Lastly, we repeat our analysis for returns, volatility, and trading volume using (i) an early sample period from January 2015 to December 2019, (ii) the COVID-19 and Russian-Ukraine war periods, and (iii) cryptocurrency ticker symbols as search keywords in place of cryptocurrency names. Although the number of observations is significantly reduced in some cases, the results are qualitatively unchanged.

To our knowledge, this is the first study examining how Google search volume affects the cross-section of cryptocurrency returns and trading activities, thus providing fresh evidence on a new asset class other than the traditional stock markets (Adachi et al., 2017; Takeda and Wakao, 2014, Joseph et al., 2011, Da et al., 2011). Our paper is also related to a large literature examining the predictability of online activities on future cryptocurrency returns (Kristoufek, 2013; Bleher and Dimpfl, 2019; Nasir et al., 2019; Dastgir et al., 2019; Panagiotidis et al., 2019, Lin, 2021, Shen et al., 2019). While our paper is closest to Smales (2022), it differs in several ways.

First, while Smales (2022) focuses on the contemporaneous relationship between investor attention and cryptocurrency returns and only briefly examines the lead-lag effect, our paper intensively studies the predictability and employs a battery of robustness tests to verify our results. Second, Smales (2022) only considers the 20 largest cryptocurrencies, while we extend the analysis scope by considering all tradable cryptocurrencies whose data are available on coinmarketcap.com. Further, we include all active and defunct cryptocurrencies in our sample to alleviate the survivorship bias, which might present if only several large cryptocurrencies are considered. The extensive set of cryptocurrencies allows us to examine the relationship in cohorts of different sizes. We find that the predictability of investor attention on cryptocurrency returns also exists among smaller cryptocurrencies, where the magnitude of the effect is less significant than that in larger cryptocurrencies. Third, Smales (2022) controls for market-wide variables such as economic policy uncertainty (EPU), cryptocurrency market uncertainty (UCRY) and financial market uncertainty (VIX). Our paper is different as it follows the standards in the asset pricing literature to control for asset-specific characteristics that have been shown to have predictive power on future returns, including beta, size, momentum, price, extreme past returns (MAX), illiquidity, idiosyncratic volatility, and coskewness, thus ruling out the possibility that those confounding factors drive the observed relationship.

Our paper also distinguishes itself from prior related literature through its methodology and sample size. Earlier studies, including those by Bleher and Dimpfl (2019), Dastgir et al. (2019), Lin (2021), and Shen et al. (2019), examine the relationship between investor attention and cryptocurrency returns and trading activities in a *time-series* setting and only focus on a few large cryptocurrencies. Smales (2022), while adopting a panel dataset for 20 cryptocurrencies over 7.5 years, only controls for cryptocurrency (cross-sectional) fixed effects in the panel data regression. The results thus only reveal the time-series relation, which has been documented in other studies. Our paper is different as we study the predictability of *all* tradable cryptocurrencies in a cross-sectional setting. This is important for two reasons. First, the cross-sectional and the time-series relations can be totally different in the same sample. The simulation in Appendix C illustrates this independence, where the simulated sample exhibits a positive time-series relation between variables X and Y but a negative cross-sectional relation. Consequently, while prior studies demonstrate a positive time-series relation between investor attention and subsequent cryptocurrency returns and trading activities, the cross-sectional relation is unknown *a priori*, at least within the context of the cryptocurrency market. Hence, it presents a research question worthy of further investigation. Second, while the trading strategy that relies on time-series predictability requires investors to take on a time-varying net long investment in risky assets, the cross-sectional strategy allows them to construct a zero-cost long-short portfolio (Goyal and Jegadeesh, 2018). Thus, our paper provides important implications on portfolio constructions and asset allocations, a dimension lacking in prior studies relying solely on time-series models.

The remaining paper is organized as follows. Section 2 describes the data and the main variables. Section 3 examines the cross-sectional relation between Google search volume and subsequent cryptocurrency returns. Section 4 presents the results of additional analysis, where we conduct a battery of robustness tests and extend the analysis to longer holding periods and cryptocurrency volatility and trading volume. Section 5 concludes.

2. Data and variable construction

2.1. Data source

We source the weekly search volume data from Google Trends over a five-year period from January 2018 to December 2022. While a longer history can be obtained if we request the data at a monthly frequency, Google Trends only returns a maximum of five years of data for a weekly frequency request. As Google Trends standardizes the time series within the period, analysis using more than five years of weekly data must first concatenate search data from different downloads using overlapping periods (Bleher and Dimpfl, 2022). Since Google Trends' standardization procedure is unknown, the concatenation process is only approximate. Thus, we only use data from one download (per cryptocurrency) for consistency reasons.[7]

Investors may search for a cryptocurrency in Google Trends using either its common name or its ticker symbol (for example, either "bitcoin" or "BTC" for Bitcoin). However, since cryptocurrencies are traded globally on multiple exchanges, different ticker symbols are expected to be used on different exchanges for one cryptocurrency (for example, XBT and BTC for Bitcoin). Further, cryptocurrencies are usually referred to by names instead of ticker symbols.[8] Thus, our main analysis uses the cryptocurrency names as the search keywords.[9] Cryptocurrencies without Google search volume data are excluded from the sample.

Data on the price, trading volume, and market capitalization of cryptocurrencies are sourced from coinmarketcap.com, a leading source of cryptocurrency trading data that has been used widely in prior related studies (Liu et al., 2022; Härdle et al., 2020; 2021). We include active and defunct cryptocurrencies to alleviate survivor bias and exclude cryptocurrencies with a market capitalization below $1 million at the end of each week. We further require a six-month wait time from the first date the cryptocurrency appears on the coinmarketcap.com database to include it in the sample. The final dataset contains 2150 cryptocurrencies, with an average of 550 cryptocurrencies per week.

[6] As small cryptocurrencies are far less known, when investors search for them, they may simply want to learn more about them without a clear intention to buy.

[7] To further demonstrate the robustness of our results, we also conduct our analysis using five-year weekly data from 2015 to 2019.

[8] For example, as of March 21, 2023, Google search returns 387 million results for "bitcoin", while the number of results for "BTC" is 167 million and for "XBT" is 3.1 million.

[9] As a robustness test, we also use ticker symbols and obtain similar results.

2.2. Construction of variables

Following Da et al. (2011), investor attention is proxied by the abnormal search volume index (*ASVI*), which is calculated as the log difference between the current SVI and the median of SVI over the last eight weeks:

$$ASVI_{i,t} = \log\left[\frac{SVI_{i,t}}{median(SVI_{i,t-1}, \ldots, SVI_{i,t-8})}\right]$$

where $SVI_{i,t}$ is the Google search volume for cryptocurrency i on week t.[10] A cryptocurrency with a higher *ASVI* is regarded as grabbing more (less) attention than its peers in the same period.

We also control for other cryptocurrency characteristics that may affect the relation between *ASVI* and future cryptocurrency returns. We first control for market beta (*Beta*), which is calculated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days before the end of each week.[11] Then, idiosyncratic volatility (*Ivol*) is calculated as the standard deviation of the residuals from the regression. We further include the natural logarithm of market capitalization (*Size*) and close price (*Price*) at the end of each week and returns in the previous week (*Mom*) to respectively control for size, price, and momentum effects (Liu et al., 2022). In addition, similar to Li et al. (2021) and Ozdamar et al. (2021), *MAX* is calculated as the average of the five highest daily returns in the previous 30 trading days. Following Amihud (2002), the illiquidity measure (*Illiq*) is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. We also control for co-skewness, which is the coefficient of the squared excess market return in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window before the end of each week (Harvey and Siddique, 2000).

Panel A of Table 1 reports the descriptive statistics of *ASVI* and other cryptocurrency characteristics, and Panel B of Table 1 presents their correlation matrix. *ASVI* has a mean of 0.061 and a median of 0. Among other cryptocurrency characteristics, *Beta*, *Size*, and *Prc* have significant negative correlations with *ASVI*, while *Mom, MAX, Illiq*, and *Ivol* are positively correlated with *ASVI*.

3. Empirical results

3.1. Portfolio analysis

3.1.1. Univariate portfolio analysis

We first examine the relation between Google search volume and future cryptocurrency returns in the univariate portfolio analysis. Specifically, at the end of each week, we sort cryptocurrencies into quintiles based on their *ASVI* during the week and calculate average returns in the subsequent week for each portfolio. We calculate both value-weighted and equal-weighted portfolio returns. Table 2 reports the results, in which portfolio "Low" ("High") comprises cryptocurrencies with the lowest (highest) *ASVI* during the week. "High – Low" illustrates the difference in returns between the two portfolios, i.e., zero-investment strategy profits that investors can earn by buying cryptocurrencies with the highest *ASVI* and selling cryptocurrencies with the lowest *ASVI* in the previous week. Columns (1) and (3) respectively report the time-series average of equal-weighted and value-weighted portfolio returns, with corresponding t-statistics in parentheses. In addition to raw returns, in columns (2) and (4), we report risk-adjusted returns, i.e., the portfolios' three-factor alphas calculated following Liu et al. (2022).[12] The results show that all return measures increase monotonically when we move from the lowest *ASVI* portfolio to the highest. The differences in returns of portfolios "High" and "Low" are positive and statistically significant at the 1 % level. Thus, the results support the positive cross-sectional relation between investor attention and subsequent cryptocurrency returns and are consistent with prior findings on stock markets (Joseph et al., 2011; Da et al., 2011).

3.1.2. Bivariate portfolio analysis

One potential issue with the univariate analysis is that the observed positive relation between *ASVI* and future returns is driven by other confounding characteristics related to cryptocurrency investors' attention. For example, the portfolio analysis presented in Appendix A shows that cryptocurrencies grabbing more abnormal attention (i.e., portfolios with higher *ASVI*) generally exhibit lower market risk (*Beta*), lower market value (*Size*) and lower price (*Prc*) but have higher past returns (*Mom*), past extreme returns (*MAX*) and idiosyncratic volatility (*Ivol*) and are more illiquid (*Illiq*).[13] We address this confounding effect issue by conducting a double-sorted portfolio analysis. Specifically, cryptocurrencies at the end of each week are divided into terciles based on a given characteristic. Then, cryptocurrencies within each characteristic tercile are further sorted into terciles based on *ASVI*, resulting in 3×3 double-sorted portfolios for each characteristic.[14] In the final step, we combine all cryptocurrencies in the three lowest *ASVI* portfolios to form portfolio "Low". Likewise, portfolios "Mid" and "High" comprise cryptocurrencies in the three middle and the three highest *ASVI* portfolios. By construction, the three *ASVI* portfolios, on average, have the same level of the given characteristic and different levels of *ASVI*, allowing the possible confounding effect of the characteristic on the relation between *ASVI* and portfolio returns to be eliminated. The double-sorting process is visualised in Appendix B. Table 3 presents the value-weighted average returns and alphas of portfolios double-sorted based on *ASVI* and various characteristics.[15] Irrespective of cryptocurrency characteristics employed in the double sorting, portfolios with the lowest *ASVI* yield smaller one-week-ahead returns and alphas than portfolios with the highest *ASVI*, and the differences are statistically significant at a 1 % level. The results suggest that the examined cryptocurrency characteristics are not the confounding factors driving the positive relation between *ASVI* and subsequent returns.

3.2. Fama-MacBeth regression analysis

As discussed in Bali et al. (2011), while the portfolio analysis has the advantage of being non-parametric and thus does not rely on the assumption of a linear relationship between variables, it omits a large amount of information in the cross-section. In addition, it does not allow control of multiple confounding factors simultaneously. In this section, we additionally employ Fama and Macbeth (1973) regression to

[10] As a result of variable construction, for a given cryptocurrency, weeks with a Google search volume of zero or with the median of Google search volume over the last five weeks of zero are excluded from the sample.

[11] We calculate excess returns as the difference between normal returns and the risk-free rate. Daily and weekly risk-free rates are proxied by the one-month Treasury bill rate, downloaded from Ken French's data library at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

[12] Specifically, alpha is calculated as the intercept in the regression of the *ASVI* portfolio's weekly returns on value-weighted market returns, small-minus-big portfolio returns, and winner-minus-loser portfolio returns. Please refer to Liu et al. (2022) for more details on the construction of these factors.

[13] The relations are somehow inconsistent with the ones presented in the correlation matrix (Panel B of Table 1). The reason is that the portfolio analysis only reveals the cross-sectional relation between variables, while the correlation matrix uses the full sample, thus including both cross-sectional and time-series relations.

[14] While we sort cryptocurrencies into quintiles in the univariate analysis, a limited number of cryptocurrencies in the early sample period does not allow a double sort into 5×5 portfolios.

[15] The results for equal-weighted returns and alphas are qualitatively similar and thus are untabulated for brevity.

Table 1
Descriptive Statistics. Panel A presents the descriptive statistics of variables used in this study, and Panel B displays their correlation matrix. *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week's Google Search Volume on the cryptocurrency name and its median over the last eight weeks. *Beta* is estimated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days prior to the end of the week. *Ivol* refers to the idiosyncratic volatility of the cryptocurrency and is calculated as the standard deviation of the residuals from the regression. *Size* and *Price* are respectively the natural log of market capitalization and close price of the cryptocurrency at the end of the week. *Mom* is the cryptocurrency return in the previous week. *MAX* is the average of the five highest daily returns in the previous 30 trading days. *Illiq* is Amihud (2002) illiquidity which is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. *Coskew* refers to co-skewness, which is calculated following Harvey and Siddique (2000) as the coefficient of the squared term in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window before the end of the week.

Panel A: Summary statistics of ASVI and other cryptocurrency characteristics

	Mean	*Median*	*Std.Dev*	*Skewness*	*Kurtosis*	*P25*	*P75*
ASVI	0.061	0.000	0.473	1.162	5.229	−0.125	0.182
Beta	0.981	1.017	0.432	−0.323	0.059	0.732	1.266
Size	17.113	16.805	2.158	0.696	0.013	15.418	18.441
Mom	−0.010	−0.013	0.250	3.021	87.189	−0.119	0.079
Prc	−1.992	−2.003	3.294	0.027	0.424	−4.094	0.001
MAX	0.120	0.097	0.083	1.602	2.855	0.064	0.152
$Illiq \times 10^6$	408.985	0.504	2714.976	8.078	66.366	0.054	4.844
Ivol	0.073	0.064	0.041	1.140	1.450	0.043	0.093
Coskew	−1.840	−1.532	4.437	−0.356	2.564	−3.855	0.278

Panel B: Correlation matrix

	ASVI	*Beta*	*Size*	*Mom*	*Prc*	*MAX*	*Illiq*	*Ivol*	*Coskew*
ASVI	1								
Beta	−0.034***	1							
Size	−0.062***	0.130***	1						
Mom	0.084***	−0.010	0.041***	1					
Prc	−0.037***	0.056**	0.490***	0.025***	1				
MAX	0.072***	0.099***	−0.273***	0.195***	−0.166***	1			
Illiq	0.018***	−0.037***	−0.164***	0.053***	−0.052***	0.286***	1		
Ivol	0.030***	0.001	−0.447***	0.065***	−0.264***	0.751***	0.336***	1	
Coskew	−0.009	0.085***	0.030**	0.005	0.073***	0.005	0.017	−0.032**	1

Table 2
Returns of single-sorted portfolios. At the end of each week from 2018 to 2022, cryptocurrencies are sorted into quintiles based on *ASVI* in that week. The low (high) portfolio is the quintile that contains cryptocurrencies with the lowest (highest) level of *ASVI*. *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week Google Search Volume on the cryptocurrency name and its median over the last eight weeks. The table reports each portfolio's one-week-ahead equal-weighted and value-weighted returns and three-factor alphas. "High – Low" denotes the difference between the portfolios with the highest and the lowest *ASVI*. Newey and West (1987) adjusted t-statistics are reported in parentheses. ***, ***, and * denote significance at 1 %, 5 %, and 10 %, respectively.

	Equal-weighted portfolios		**Value-weighted portfolios**	
	Average returns (1)	***Alpha*** (2)	***Average returns*** (3)	***Alpha*** (4)
Low	−0.034***	−0.035***	−0.026***	−0.024***
	(−4.708)	(3.904)	(−3.637)	(2.771)
2	−0.024***	−0.028***	−0.013*	−0.015*
	(−3.256)	(2.979)	(−1.870)	(1.749)
3	−0.020***	−0.024	−0.013*	−0.011
	(−2.968)	(2.678)	(−1.884)	(1.306)
4	−0.020***	−0.022**	−0.002	−0.002
	(−2.720)	(2.508)	(−0.246)	(−0.187)
High	−0.006	−0.015	0.016*	0.009
	(−0.884)	(1.632)	(1.775)	(0.788)
High - Low	0.028***	0.021***	0.042***	0.033***
	(13.482)	(8.709)	(7.187)	(4.627)

compensate for these drawbacks. Specifically, we first run cross-sectional regressions of one-week-ahead returns on *ASVI*, controlling for other cryptocurrency characteristics used in the previous double-sorted portfolio analysis (i.e., *Beta, Size, Mom, Price, MAX, Illiq, Ivol, Coskew*). We then calculate the time-series average of the slope coefficients, along with their corresponding Newey-West adjusted t-statistics. Table 4 shows that in all model specifications, the coefficients of *ASVI* are all significantly positive at a 1 % level, implying that the effect of *ASVI* on subsequent cryptocurrency returns is not subsumed by other cryptocurrency characteristics. The effect is also economically significant. For example, the estimated coefficient of *ASVI* is 0.03 in the full model specification (column (8)), indicating that one standard deviation increase in *ASVI* (0.473) leads to an increase of 1.5 percentage points in the next week's price. Therefore, the Fama and Macbeth (1973) regression is consistent with our main findings using univariate and bivariate portfolio analysis.

4. Additional analysis

4.1. Effects of cryptocurrency size

Prior studies document that the magnitude of the effect of Google search volume on subsequent stock returns depends on firm size. For instance, Da et al. (2011) and Takeda and Wakao (2014) examine the U. S. and Japanese stock markets and show that the relationship is more significant for small stocks. To examine whether a similar result holds in the cryptocurrency market, we run the Fama-MacBeth regression with the interaction term between *ASVI* and the natural log of cryptocurrency market value (*Size*) as an independent variable. Results in column (1) of Table 5 show a positive and statistically significant coefficient of *ASVI*×*Size*, indicating that in contrast to stock markets, the original results are stronger among larger cryptocurrencies. For ease of interpretation regarding the economic significance of the difference, we further split the full sample into three size subsamples based on terciles of cryptocurrency market value at the end of each week and ran the Fama-MacBeth regression for each subsample. While the coefficients of *ASVI* are significantly positive in all three subsamples, their magnitudes decrease monotonically when we move from the Large subsample (column (2)) to the Small subsample (column (4)). The magnitude of the coefficients of *ASVI* shows that the effect is about two times stronger among large cryptocurrencies compared to the small counterparts (0.042 vs 0.020).

What explains the difference between the stock market and the cryptocurrency market regarding the size effect? The negative size effect found on the stock market is attributed to the fact that small stocks are

Table 3
Returns of double-sorted portfolios. At the end of each week from 2018 to 2022, cryptocurrencies are sorted into tercile based on a given characteristic (*Beta, Size, Mom, Prc, Max, Illiq, Ivol,* and *Coskew)*. We then form tercile portfolios based on *ASVI* within each characteristic tercile. Portfolio High (Low) is the combined portfolio of cryptocurrencies with the highest (lowest) *ASVI* in each characteristic tercile. The table reports the one-week-ahead value-weighted returns (Panel A) and three-factor alphas (Panel B) of each combined *ASVI* portfolio and the difference between the High and Low portfolios in the last two rows. The column names represent the corresponding characteristics used in the double-sorting process. Newey and West (1987) adjusted t-statistics are reported in parentheses. ***, ***, and * denote significance at 1 %, 5 %, and 10 %, respectively.

	Double-sorted based on:							
	Beta (1)	*Size* (2)	*Mom* (3)	*Price* (4)	*MAX* (5)	*Illiq* (6)	*Ivol* (7)	*Coskew* (8)
Panel A: Value-weighted average returns on double-sorted portfolios								
Low	−0.023***	−0.022***	−0.025***	−0.027***	−0.024***	−0.026**	−0.026***	−0.025***
	(−3.106)	(−2.968)	(−3.026)	(−2.897)	(−2.871)	(−2.387)	(−2.844)	(−3.043)
Mid	−0.02*	−0.022*	−0.013*	−0.011	−0.014**	−0.014*	−0.012*	−0.012*
	(−1.927)	(−1.831)	(−1.784)	(−1.529)	(−2.102)	(−1.859)	(−1.73)	(−1.681)
High	0.011	0.007	0.011	0.007	0.011	0.009	0.009	0.008
	(1.253)	(0.804)	(1.244)	(0.760)	(1.236	(1.117)	(1.093)	(0.956)
High – Low	0.034***	0.028***	0.036***	0.033***	0.034***	0.035***	0.035***	0.034***
	(4.922)	(4.564)	(5.058)	(3.872)	(5.282)	(4.041)	(4.315)	(4.322)
Panel B: Value-weighted alphas on double-sorted portfolios								
Low	−0.024***	−0.021**	−0.027***	−0.029**	−0.026***	−0.027**	−0.029**	−0.027***
	(−2.823)	(−2.532)	(−2.765)	(−2.404)	(−2.898)	(−2.126)	(−2.452)	(−2.738)
Mid	−0.021*	−0.025*	−0.013**	−0.010	−0.010	−0.014*	−0.013**	−0.010
	(−1.716)	(−1.781)	(−2.033)	(−1.463)	(−1.568)	(−1.966)	(−1.97)	(−1.372)
High	0.007	0.002	0.009	0.004	0.008	0.005	0.006	0.005
	(0.915)	(0.256)	(1.170)	(0.477)	(1.020)	(0.634)	(0.746)	(0.637)
High – Low	0.031***	0.023***	0.036***	0.033***	0.034***	0.032***	0.034***	0.032***
	(3.609)	(3.143)	(3.869)	(2.944)	(3.892)	(2.712)	(3.044)	(3.414)

Table 4
Fama-Macbeth regression. This table reports results from Fama and Macbeth (1973) regressions of one-week-ahead returns on *ASVI* and various cryptocurrency characteristics, including *Beta, Size, Mom, Prc, Max, Illiq, Ivol,* and *Coskew*. *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week's Google Search Volume on the cryptocurrency name and its median over the last eight weeks. *Beta* is estimated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days prior to the end of the week. *Ivol* refers to the idiosyncratic volatility of the cryptocurrency and is calculated as the standard deviation of the residuals from the regression. *Size* and *Price* are respectively the natural log of market capitalization and close price of the cryptocurrency at the end of the week. *Mom* is the cryptocurrency return in the previous week. *MAX* is the average of the five highest daily returns in the previous 30 trading days. *Illiq* is Amihud (2002) illiquidity which is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. *Coskew* refers to co-skewness which is calculated following Harvey and Siddique (2000) as the coefficient of the squared term in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window prior to the end of the week. Newey and West (1987) adjusted t-statistics are reported in parentheses. ***, ***, and * denote significance at 1 %, 5 %, and 10 %, respectively.

	Dependent variable: One-week-ahead returns							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ASVI	0.025***	0.026***	0.030***	0.030***	0.031***	0.031***	0.030***	0.030***
	(6.765)	(7.009)	(7.596)	(7.739)	(7.822)	(7.886)	(7.464)	(7.642)
Beta	0.004	0.003	0.00001	0.0001	0.005	0.005	0.004	0.004
	(1.040)	(0.663)	(0.001)	(0.026)	(1.058)	(1.023)	(0.833)	(0.853)
Size		0.002***	0.002***	0.003***	0.0004	0.0003	−0.001	−0.001
		(3.093)	(4.031)	(4.558)	(0.655)	(0.455)	(−1.281)	(−1.186)
Mom			−0.121***	−0.122***	−0.104***	−0.101***	−0.105***	−0.106***
			(−12.144)	(−12.233)	(−10.229)	(−10.091)	(−10.297)	(−10.373)
Prc				−0.0003	−0.001	−0.001	−0.001	−0.001
				(−0.617)	(−1.019)	(−0.992)	(−1.353)	(−1.478)
MAX					−0.208***	−0.199***	−0.098***	−0.094***
					(−11.985)	(−11.109)	(−4.623)	(−4.453)
Illiq						−10.347*	−6.114	−5.335
						(−1.856)	(−1.065)	(−0.929)
Ivol							−0.291***	−0.297***
							(−6.495)	(−6.646)
Coskew								0.00001
								(0.016)
Constant	−0.028***	−0.056***	−0.068***	−0.071***	−0.015	−0.013	0.017	0.016
	(−4.514)	(−4.389)	(−4.830)	(−5.752)	(−1.048)	(−0.925)	(1.084)	(1.038)
Obs.	141,357	141,357	141,357	141,357	141,357	141,357	141,357	141,357
R^2	0.249	0.252	0.293	0.297	0.306	0.313	0.317	0.320

traded mainly by individual investors. Unlike institutional investors who usually have access to more sophisticated information services such as Reuters or Bloomberg, individual investors are more likely to rely on the Google search engine to obtain information, leading to a larger price impact of search volume on small stocks (Da et al., 2011). We postulate that such a mechanism and, thus, the negative moderating effect of size are not necessarily true for the cryptocurrency market due to its unique characteristics compared to the stock market. First, cryptocurrencies are a new asset class that is still dominated by retail investors. Although some institutional investors started including cryptocurrencies in their portfolios recently, they mostly pay attention to leading cryptocurrencies such as Bitcoin or Ethereum. Second, while institutional investors

Table 5
Effects of cryptocurrency size. Panel A reports the regression results for the full sample with the interaction of *ASVI* and *Size* (i.e., *ASVI*×*Size*) as an independent variable. *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week's Google Search Volume on the cryptocurrency name and its median over the last eight weeks. *Beta* is estimated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days prior to the end of the week. *Ivol* refers to the idiosyncratic volatility of the cryptocurrency and is calculated as the standard deviation of the residuals from the regression. *Size* and *Price* are respectively the natural log of market capitalization and close price of the cryptocurrency at the end of the week. *Mom* is the cryptocurrency return in the previous week. *MAX* is the average of the five highest daily returns in the previous 30 trading days. *Illiq* is Amihud's (2002) illiquidity, which is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. *Coskew* refers to co-skewness, which is calculated following Harvey and Siddique (2000) as the coefficient of the squared term in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window prior to the end of the week. Panel B reports the regression results without the interaction term for three subsamples split by size. Specifically, at the end of each week, cryptocurrencies are sorted into terciles based on market capitalization. *Small, Medium,* and *Large* subsamples include cryptocurrencies in the lowest, middle, and highest size terciles.

	Panel A: Full sample	***Panel B: Subsample divided by size***		
		Large	***Medium***	***Small***
	(1)	(2)	(3)	(3
ASVI	−0.072***	0.042***	0.026***	0.020***
	(−2.939)	(7.614)	(7.100)	(4.809)
ASVI×*Size*	0.006***			
	(3.928)			
Beta	−0.001*	−0.001	−0.0003	−0.002
	(−1.690)	(−0.738)	(−0.111)	(−0.741)
Size	0.004	0.010	−0.001	0.005
	(0.796)	(1.595)	(−0.151)	(0.728)
Mom	−0.106***	−0.024	−0.059***	−0.129***
	(−10.444)	(−1.411)	(−4.176)	(−9.061)
Prc	−0.001	−0.001*	−0.001	−0.001
	(−1.514)	(−1.781)	(−0.983)	(−1.508)
Max	−0.098***	−0.215***	−0.079***	−0.087***
	(−4.629)	(−5.651)	(−2.859)	(−3.017)
Illiq	−5.144	−21,074.880	−164.130	−3.479
	(−0.914)	(−1.444)	(−0.967)	(−0.578)
Ivol	−0.291***	−0.320***	−0.312***	−0.200***
	(−6.707)	(−4.355)	(−4.836)	(−3.413)
Coskew	−0.00002	−0.0001	0.0004	−0.001
	(−0.047)	(−0.114)	(0.792)	(−1.003)
Constant	0.026	0.018	0.013	0.022
	(1.481)	(0.919)	(0.318)	(0.565)
Obs.	141,357	47,115	46,957	47,115
R^2	0.325	0.550	0.390	0.280

Table 6
Longer holding periods. This table reports results from Fama and MacBeth's (1973) regressions of long-term returns on *ASVI* and various cryptocurrency characteristics, including *Beta, Size, Mom, Prc, Max, Illiq, Ivol,* and *Coskew*. Long-term returns include two-, three-, and four-week-ahead returns (columns (1) – (3)) and returns from week 5 to week 52 (column (4)). *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week's Google Search Volume on the cryptocurrency name and its median over the last eight weeks. *Beta* is estimated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days prior to the end of the week. *Ivol* refers to the idiosyncratic volatility of the cryptocurrency and is calculated as the standard deviation of the residuals from the regression. *Size* and *Price* are respectively the natural log of market capitalization and close price of the cryptocurrency at the end of the week. *Mom* is the cryptocurrency return in the previous week. *MAX* is the average of the five highest daily returns in the previous 30 trading days. *Illiq* is Amihud (2002) illiquidity which is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. *Coskew* refers to co-skewness, which is calculated following Harvey and Siddique (2000) as the coefficient of the squared term in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window prior to the end of the week. Newey and West (1987) adjusted t-statistics are reported in parentheses. ***, ***, and * denote significance at 1 %, 5 %, and 10 %, respectively.

	Dependent variable:			
	Ret_{t+2}	Ret_{t+3}	Ret_{t+4}	$Ret_{[t+5,t+52]}$
	(1)	(2)	(3)	(4)
ASVI	0.001	−0.001	0.001	−0.019***
	(0.557)	(−1.232)	(0.578)	(−11.411)
Beta	0.002	0.003	0.002	0.080***
	(0.435)	(0.694)	(0.536)	(16.922)
Size	0.0004	−0.0002	0.0002	0.062***
	(0.777)	(−0.321)	(0.469)	(117.792)
Mom	0.0004	−0.002	−0.0001	−0.039***
	(0.038)	(−0.202)	(−0.009)	(−6.781)
Prc	−0.001**	−0.001*	−0.001**	−0.021***
	(−2.524)	(−1.690)	(−2.211)	(−44.588)
MAX	−0.066***	−0.045***	−0.049***	−0.001
	(−4.119)	(−2.581)	(−2.745)	(−0.066)
Illiq	9.277**	11.239**	9.233*	331.598***
	(2.224)	(2.505)	(1.911)	(51.700)
Ivol	−0.178***	−0.205***	−0.167***	−3.386***
	(−5.162)	(−6.990)	(−4.769)	(−83.831)
Coskew	0.001**	−0.0002	0.0001	−0.010***
	(1.995)	(−0.517)	(0.373)	(−27.128)
Constant	−0.006	0.002	−0.006	−1.387***
	(−0.597)	(0.217)	(−0.587)	(−108.576)
Obs.	140,581	139,761	138,957	95,217
R^2	0.307	0.291	0.290	0.492

have advantages over retail investors in collecting and analysing stock-related information such as companies' fundamentals, operations or reports, such advantages diminish significantly in the cryptocurrency market due to the transparency of blockchain and the lack of fundamentals. Finally, since large cryptocurrencies are more well-known than small ones when investors search for the latter, they may simply want to learn more about them without an intention to buy, leading to a weaker relation between search volume and subsequent returns among smaller cryptocurrencies.

4.2. Longer holding periods

We next examine the predictive power of *ASVI* for long-term returns by running the Fama-MacBeth regressions with four alternative dependent variables, including two-, three-, and four-week ahead returns (Ret_{t+2}, Ret_{t+3}, Ret_{t+4} respectively), and returns from week 5 to week 52 ($Ret_{[t+5,\ t+52]}$). Results reported in Table 6 show that while coefficients of *ASVI* for Ret_{t+2}, Ret_{t+3} and Ret_{t+4} (columns (1) – (3)) are insignificant, that of $Ret_{[t+5,\ t+52]}$ (column (4)) is significantly negative at a 1 % level, indicating a price reversal in the long run. However, the magnitude of reversal is smaller than the initial increase (i.e., −0.019 versus 0.030 (shown in Table 4)). The result is partly inconsistent with Da et al. (2011) on the U.S. stock market, which exhibits a full reversal after one year. We attribute this difference to the fact that, unlike stocks, cryptocurrency's intrinsic value is far more difficult to determine due to the lack of fundamentals, and prices are strongly affected by investors' belief in future appreciation and network extension (Liu and Tsyvinski, 2021). Thus, it is intuitive that investor attention can permanently shift cryptocurrency prices.

4.3. ASVI and cross-sectional variation in volatility and trading volume

As a natural extension of the analysis, we next examine whether Google search volume affects future return volatility and trading volume. First, we estimate weekly cryptocurrency volatility as the square root of average squared daily returns during the week. We calculate the weekly trading volume for each cryptocurrency as the average daily trading volume during the week. As trading volume tends to increase over time, we focus on abnormal trading volume, calculated as the

difference between weekly trading volume and its average in the last 52 weeks, scaled by its standard deviation over the same period. Table 7 shows positive and statistically significant coefficients of *ASVI* for both volatility and trading volume regression, indicating that cryptocurrencies grabbing more attention from investors subsequently are traded more and exhibit higher volatility. The results lend further support to the attention-induced price pressure hypothesis.

4.4. Using cryptocurrency symbols as search terms

In the main analysis, we use a cryptocurrency's name to retrieve its Google search volume. In this section, we test the robustness of our results using ticker symbols as the search keywords. While different exchanges may use different symbols for the same cryptocurrency, we opt to use the symbols from coinmarketcap.com, arguably the most common source of cryptocurrency price and volume data. We then recalculate *ASVI* based on search volume on symbols and rerun the baseline Fama-MacBeth regression for returns, volatility, and trading volume. Table 8 shows that the coefficients of *ASVI* are significantly positive in all three regressions, indicating that our original results are not affected by the variation in the search keywords.

4.5. Does the sample period matter?

In this section, we further test whether our results are robust to different sample periods. First, we examine a period in which cryptocurrencies have not attracted the current level of investor attention. Specifically, we repeat our analysis utilising an alternative five-year

Table 7
ASVI and cross-sectional variation in volatility and trading volume. This table reports results from Fama and Macbeth (1973) regressions of one-week-ahead volatility and trading volume on *ASVI* and various cryptocurrency characteristics, including *Beta, Size, Mom, Prc, Max, Illiq, Ivol, Coskew*, past return and past trading volume. *Volatility* is calculated as the square root of average squared daily returns during the week. *Volume* is the difference between weekly trading volume and its average in the last 52 weeks, scaled by its standard deviation over the same period. *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week's Google Search Volume on the cryptocurrency name and its median over the last eight weeks. *Beta* is estimated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days prior to the end of the week. *Ivol* refers to the idiosyncratic volatility of the cryptocurrency and is calculated as the standard deviation of the residuals from the regression. *Size* and *Price* are respectively the natural log of market capitalization and close price of the cryptocurrency at the end of the week. *Mom* is the cryptocurrency return in the previous week. *MAX* is the average of the five highest daily returns in the previous 30 trading days. *Illiq* is Amihud's (2002) illiquidity, which is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. *Coskew* refers to co-skewness, which is calculated following Harvey and Siddique (2000) as the coefficient of the squared term in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window prior to the end of the week. Newey and West (1987) adjusted t-statistics are reported in parentheses. ***, ***, and * denote significance at 1 %, 5 %, and 10 %, respectively.

	Dependent variable:			
	$Volatility_{t+1}$		$Volume_{t+1}$	
	(1)	(2)	(3)	(4)
ASVI	0.025***	0.015***	0.487***	0.381***
	(11.131)	(7.607)	(5.179)	(5.317)
Beta		0.003		−0.065
		(0.740)		(−1.354)
Size		−0.0003		0.054***
		(−0.549)		(5.124)
Mom		−0.055***		0.573***
		(−4.915)		(3.902)
Prc		−0.001***		0.004
		(−5.571)		(0.440)
MAX		0.074**		0.634*
		(2.256)		(1.863)
Illiq		171.887**		−724.132*
		(2.546)		(−1.749)
Ivol		0.693***		0.462
		(17.640)		(0.856)
Coskew		−0.001		−0.006
		(−1.176)		(−1.349)
Volatility		0.419***		−0.334***
		(10.453)		(−4.275)
Volume		−0.005***		0.528***
		(−5.003)		(16.902)
Constant	0.179***	0.044***	0.487***	0.381***
	(29.702)	(3.733)	(5.179)	(5.317)
Obs.	89,330	89,330	89,330	89,330
R^2	0.040	0.662	0.066	0.157

Table 8
ASVI using cryptocurrency symbols. This table reports results from Fama and Macbeth (1973) regressions of one-week-ahead returns (column (1)), volatility (column (2)), and trading volume (column (3)) on *ASVI* and various cryptocurrency characteristics, including *Beta, Size, Mom, Prc, Max, Illiq, Ivol, Coskew*, past return and past trading volume. *Volatility* is calculated as the square root of average squared daily returns during the week. *Volume* is the difference between weekly trading volume and its average in the last 52 weeks, scaled by its standard deviation over the same period. *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week's Google Search Volume on the cryptocurrency symbols and its median over the last eight weeks. *Beta* is estimated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days before the end of the week. *Ivol* refers to the idiosyncratic volatility of the cryptocurrency and is calculated as the standard deviation of the residuals from the regression. *Size* and *Price* are respectively the natural log of market capitalization and close price of the cryptocurrency at the end of the week. *Mom* is the cryptocurrency return in the previous week. *MAX* is the average of the five highest daily returns in the previous 30 trading days. *Illiq* is Amihud (2002) illiquidity which is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. *Coskew* refers to co-skewness, which is calculated following Harvey and Siddique (2000) as the coefficient of the squared term in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window prior to the end of the week. Newey and West (1987) adjusted t-statistics are reported in parentheses. ***, ***, and * denote significance at 1 %, 5 %, and 10 %, respectively.

	Dependent variable:		
	Ret_{t+1}	$Volatility_{t+1}$	$Volume_{t+1}$
	(1)	(2)	(3)
ASVI	0.025***	0.019***	0.500***
	(6.935)	(6.718)	(4.928)
Beta	0.003	0.007***	−0.084*
	(0.708)	(3.003)	(−1.947)
Size	0.0002	−0.001***	0.054***
	(0.324)	(−2.747)	(4.995)
Mom	−0.071***	−0.047***	0.570***
	(−14.460)	(−5.448)	(4.412)
Prc	−0.001*	−0.002***	−0.001
	(−1.763)	(−6.684)	(−0.121)
MAX	−0.096***	0.140***	0.479*
	(−7.398)	(5.534)	(1.783)
Illiq	0.028	41.851**	−411.422**
	(0.089)	(2.496)	(−2.314)
Ivol	−0.167***	0.821***	0.506
	(−5.840)	(19.682)	(1.254)
Coskew	0.0003	−0.0001	−0.004
	(1.161)	(−0.278)	(−1.102)
Volatility		0.363***	−0.220***
		(13.882)	(−2.644)
Volume		−0.005***	0.507***
		(−5.959)	(16.284)
Constant	−0.005	0.052***	−0.777***
	(−0.376)	(5.515)	(−4.985)
Obs.	186,893	117,355	117,355
R^2	0.343	0.513	0.124

sample period from January 2015 to December 2019. As expected, the sample size is reduced significantly, with 862 cryptocurrencies and an average of 181 cryptocurrencies per week. Second, we examine the relationship between the COVID-19 pandemic and the Russian-Ukraine war periods, which greatly increased uncertainty in the financial markets and triggered a significant sell-off in both financial and cryptocurrency markets. We follow Pástor and Vorsatz (2020) to use February 20, 2020, as the starting date of the COVID-19 period and assume December 31, 2021, as the end date.[16] The Russian-Ukraine war period is from February 24, 2022 (the Russian invasion starting day) to the end of our sample period (December 31, 2022). The results in Table 9 show that despite a much smaller sample size in all three alternative sample periods, the magnitude of *ASVI* coefficients and their statistical significance remain, indicating that the ASVI effect is robust to the variation of sample periods.

5. Conclusions

This paper examines the effect of investor attention, which is proxied by Google search volume, on the cross-section of cryptocurrency returns and trading activity. Using standard asset pricing tools, including portfolio analyses and Fama-MacBeth regressions, we demonstrate that cryptocurrencies associated with higher abnormal search volume subsequently exhibit higher returns, higher volatility, and higher trading volume. The results are robust to alternative sample periods and alternative search keywords, providing concrete support to the attention-induced price pressure hypothesis and consistent with prior studies on the equity market. However, we discover two findings that are distinct from the stock market. First, we find a positive moderating effect of cryptocurrency size, in which the relationship between search volume and returns is more pronounced among larger cryptocurrencies. Second, we only observe a partial reversal of the prices after the initial increase, implying that investor attention permanently shifts the cryptocurrency prices. We attribute the differences to a number of unique characteristics of cryptocurrencies compared to stocks, including (1) the dominance of retail investors, (2) the transparency of blockchain and the lack of fundamentals, and (3) the popularity of large cryptocurrencies. Our findings contribute to understanding factors affecting cryptocurrency prices and have important implications for portfolio formation and asset allocation.

CRediT authorship contribution statement

Lai Hoang: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation. **Duc Hong Vo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors have declared no conflict of interest.

Appendix A. Characteristics of portfolios sorted by *ASVI*.

At the end of each week from 2018 to 2022, cryptocurrencies are sorted into quintiles based on *ASVI* in that week. The low (high) portfolio is the quintile that contains cryptocurrencies with the lowest (highest) level of *ASVI*. The table reports the weekly average of various characteristics of the *ASVI*-sorted portfolios for each quintile. *ASVI* is the abnormal search volume index, which is calculated as the log difference between this week's Google Search Volume on the cryptocurrency name and its median over the last eight weeks. *Beta* is estimated by regressing daily excess returns of each cryptocurrency on the excess value-weighted market returns during the past 90 days prior to the end of the week. *Ivol* refers to the idiosyncratic volatility of the cryptocurrency and is calculated as the standard deviation of the residuals from the regression. *Size* and *Price* are respectively the natural log of market capitalization and close price of the cryptocurrency at the end of the week. *Mom* is the cryptocurrency return in the previous week. *MAX* is the average of the five highest daily returns in the previous 30 trading days. *Illiq* is Amihud's (2002) illiquidity, which is calculated as the average of the absolute daily returns divided by the daily dollar trading volume during the week. *Coskew* refers to co-skewness, which is calculated following Harvey and Siddique (2000) as the coefficient of the squared term in the regression of daily excess cryptocurrency returns on the daily excess market returns and the squared daily excess market returns in a 90-day window prior to the end of the week. "High – Low" denotes the difference between the portfolios with the highest and the lowest *ASVI*. t-statistics are reported in parentheses. ***, ***, and * denote significance at 1 %, 5 %, and 10 %, respectively.

	Beta	*Size*	*Mom*	*Price*	*MAX*	*Illiq*	*Ivol*	*Coskew*
Low	0.985***	16.996***	−0.031***	−2.095***	0.116***	35.612***	0.073***	−1.842***
	(97.644)	(418.195)	(−4.125)	(−48.911)	(67.736)	(19.445)	(81.389)	(−18.979)
2	0.984***	17.054***	−0.016**	−2.05***	0.117***	42.03***	0.071***	−1.925***
	(102.320)	(450.116)	(−2.078)	(−42.424)	(64.056)	(20.981)	(77.812)	(−20.537)
3	0.983***	17.061***	−0.012	−2.151***	0.117***	43.661***	0.07***	−1.891***
	(104.461)	(539.601)	(−1.502)	(−42.847)	(61.246)	(25.692)	(73.433)	(−20.546)
4	0.971***	16.941***	−0.007	−2.036***	0.12***	46.063***	0.072***	−1.804***
	(106.279	(471.215)	(−1.012)	(−41.677)	(64.617)	(25.375)	(79.413)	(−20.316)
High	0.948***	16.635***	0.016**	−2.382***	0.13***	51.951***	0.076***	−1.916***
	(109.012)	(428.655)	(2.016)	(−48.359)	(62.729)	(21.510)	(77.435)	(−20.434)
High - Low	−0.037***	−0.361***	0.048***	−0.286***	0.015***	16.339***	0.003***	−0.073*
	(−7.121)	(−9.025)	(15.795)	(−7.958)	(16.710)	(6.149)	(5.966)	(−1.707)

[16] While there are no widely accepted, quantitative definitions for the end date of the COVID-19 pandemic, Ioannidis (2022) shows that a critical threshold of 70 % of the global population having been vaccinated or infected was reached in the second half of 2021, and the death toll and intensive care unit (ICU) occupancy figures are consistent with a transition to endemicity by the end of 2021/early 2022.

Table 9
Alternative sample periods.

	Panel A: Alternative 5-year period (31/12/2014 – 31/12/2019)			Panel B: COVID-19 period. (20/03/2020 – 31/01/2022)			Panel C: Russian-Ukraine war period (01/02/2022 – 31/12/2022)		
Dependent variable	Ret_{t+1} (1)	$Volatility_{t+1}$ (2)	$Volume_{t+1}$ (3)	Ret_{t+1} (4)	$Volatility_{t+1}$ (5)	$Volume_{t+1}$ (6)	Ret_{t+1} (7)	$Volatility_{t+1}$ (8)	$Volume_{t+1}$ (9)
ASVI	0.038***	0.019***	0.572*	0.045***	0.018***	0.557***	0.016***	0.017***	0.246***
	(6.032)	(3.555)	(1.667)	(6.777)	(5.293)	(4.096)	(5.977)	(7.513)	(6.028)
Beta	−0.008	0.012**	0.424	0.019**	−0.006	−0.091	−0.002	0.012**	−0.007
	(−1.039)	(2.063)	(1.073)	(2.463)	(−1.054)	(−0.771)	(−0.157)	(2.494)	(−0.489)
Size	−0.001	−0.003**	−0.086	−0.001	0.001	0.078***	0.00001	−0.0004	−0.002
	(−0.731)	(−2.090)	(−0.773)	(−1.320)	(1.021)	(4.960)	(0.009)	(−0.595)	(−0.157)
Mom	−0.021	−0.059**	1.804	−0.095***	−0.083***	1.136***	−0.118***	−0.060***	0.182
	(−0.989)	(−2.557)	(1.582)	(−6.938)	(−3.738)	(3.697)	(−3.194)	(−2.959)	(1.501)
Prc	−0.001	−0.001*	0.006	−0.002**	−0.002***	−0.004	0.0005	−0.001**	0.009***
	(−0.611)	(−1.655)	(0.213)	(−2.180)	(−3.864)	(−0.208)	(1.238)	(−2.408)	(2.870)
MAX	−0.219***	0.162**	1.990	−0.113***	−0.039	1.403*	−0.068	0.085	0.003
	(−2.627)	(2.535)	(0.667)	(−3.381)	(−0.646)	(1.890)	(−1.182)	(1.447)	(0.007)
Illiq	−86.020	411.706	39,082	−6.294	133.6***	−314.4***	−13.784*	17.462**	−56.498**
	(−1.349)	(1.463)	(0.952)	(−0.554)	(4.403)	(−3.042)	(−1.854)	(2.078)	(−2.418)
Ivol	−0.044	0.764***	−3.148	−0.319***	0.635***	−0.085	−0.308***	0.755***	2.227**
	(−0.542)	(8.250)	(−0.716)	(−3.766)	(10.189)	(−0.080)	(−5.234)	(8.017)	(2.312)
Coskew	0.001	0.001	0.056	−0.001	−0.002	−0.012	−0.0003	−0.001***	0.006
	(0.562)	(1.067)	(1.589)	(−0.688)	(−1.534)	(−1.191)	(−0.791)	(−3.479)	(1.004)
Volatility		0.307***	−0.668		0.575***	−0.287***		0.412***	−0.313***
		(8.323)	(−0.476)		(9.234)	(−3.810)		(5.574)	(−3.896)
Volume		−0.005**	0.727***		−0.004***	0.411***		−0.006***	0.498***
		(−1.975)	(6.112)		(−3.571)	(10.638)		(−3.497)	(9.288)
Constant	0.052	0.086***	1.756	0.044**	0.016	−1.021***	−0.024	0.022	−0.248
	(1.603)	(3.879)	(0.929)	(1.963)	(0.816)	(−4.281)	(−1.290)	(1.332)	(−0.880)
Obs.	45,559	21,565	21,565	64,348	40,247	40,247	39,473	28,609	28,609
R^2	0.466	0.652	0.464	0.276	0.689	0.132	0.347	0.525	0.079

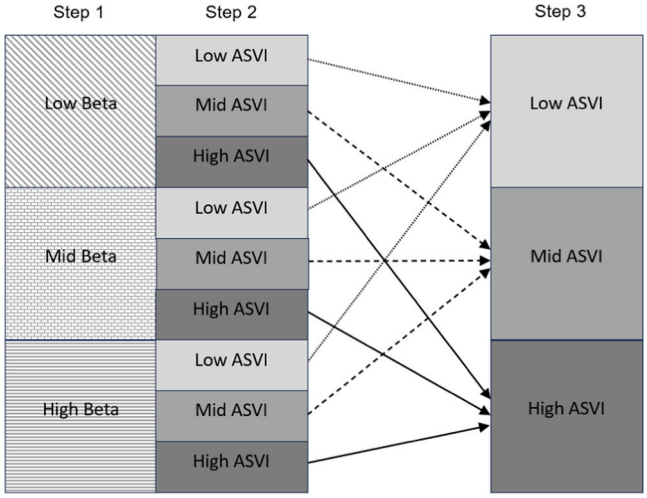
Appendix B. Portfolio double-sorting process

This Appendix illustrates the double-sorting process described in Section 3.1.2 for two variables, *Beta* and *ASVI*. The process for other cryptocurrency characteristics is similar. It is done in a three-step procedure described as follows:

Step 1: At the end of each week, cryptocurrencies are divided into terciles (low, mid, and high beta) based on its beta estimated using data up to the end of the week.

Step 2: Cryptocurrencies within each beta tercile are further sorted into terciles based on *ASVI* at the end of the week (low, mid, and high *ASVI*).

Step 3: Cryptocurrencies in the portfolios with the same level of *ASVI* are combined to form the three portfolios "Low ASVI", "Mid ASVI", and "High ASVI".



Appendix C. Simulation

We simulate variables X and Y of five assets (i = A, B, C, D, and E) over 20 periods (t = 1, 2, …, 20) using the following equations:

$$Y_i = \alpha_{ts} + \beta_{ts} X_i + \varepsilon_i$$

$$Y_t = \alpha_{cs} + \quad \beta_{cs} X_t + \epsilon_t$$

where i and t respectively denote asset and time, α_{ts} and α_{cs} are respectively time-series and cross-sectional constants, and randomly generated within [0,50] and [0,20], and ε_i and ϵ_t are white noise errors. For illustration purposes, we use $\beta_{ts} = 0.5$ and $\beta_{cs} = -0.5$. The R codes used for the simulation is available in Supplementary Data. The scatter plot of variables X and Y presented in Figure C1 shows that within the same simulated sample, the time series and cross-sectional relation are totally different. Specifically, while the relation between the two variables is negative for each asset, it is positive for each period.

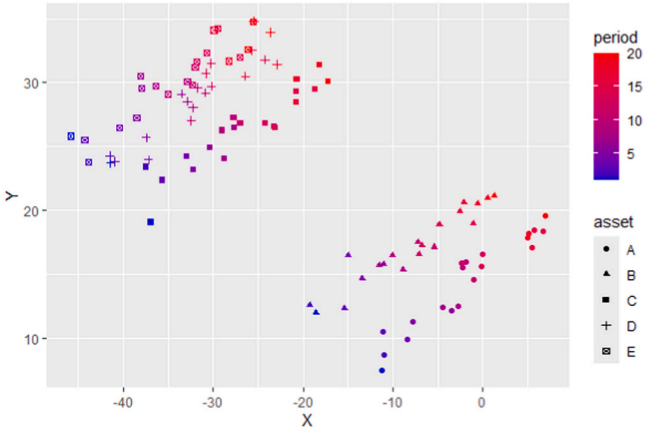


Figure C1. The scatter plot of X and Y

The regression results using different estimation methods presented in Table C1 also show that while the time-series regression for each individual asset and the panel data regression that only control for asset (cross-sectional) fixed effects reveal a negative relation between X and Y (columns (1) – (6)), the remaining three estimation methods including pool regression, panel data regression with time fixed effects and Fama-MacBeth (1973) regression reveal a positive relation (columns (7) – (9)).

Table C1
Regression results of the simulated data

	Model: $Y = a + bX + e$								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
X	0.558***	0.430***	0.448***	0.534***	0.457***	0.484***	−0.320***	−0.485***	−0.486***
	(13.579)	(12.861)	(10.386)	(10.799)	(9.753)	(24.776)	(−8.476)	(−60.118)	(−53.885)
Constant	15.367***	20.474***	38.256***	45.874***	45.783***		16.853***		13.415***
	(60.523)	(61.216)	(32.479)	(28.882)	(28.046)		(17.833)		(9.734)
Obs.	20	20	20	20	20	100	100	100	100
R^2	0.911	0.902	0.857	0.866	0.841	0.867	0.423	0.979	0.975
Adj. R^2	0.906	0.896	0.849	0.859	0.832	0.860	0.417	0.973	

Columns (1) – (5) present the OLS regression results for each asset A – E individually. Column (6) displays the results using panel data regression with only cross-sectional fixed effects. Results using pool OLS regression (without any fixed effect), panel data regression with only time fixed effects, and Fama-MacBeth regression are respectively presented in columns (7) – (8).

Appendix D. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jbef.2024.100991.

References

Adachi, Y., Masuda, M., Takeda, F., 2017. Google search intensity and its relationship to the returns and liquidity of Japanese startup stocks. Pac. -Basin Financ. J. 46, 243–257.
Ahmed, S., Grobys, K., Sapkota, N., 2020. Profitability of technical trading rules among cryptocurrencies with privacy function. Financ. Res. Lett. 35, 101495.
Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. J. Financ. Mark. 5, 31–56.
Aouadi, A., Arouri, M., Teulon, F., 2013. Investor attention and stock market activity: Evidence from France. Econ. Model. 35, 674–681.
Bali, T.G., Cakici, N., Whitelaw, R.F., 2011. Maxing out: stocks as lotteries and the cross-section of expected returns. J. Financ. Econ. 99, 427–446.
Bank, M., Larch, M., Peter, G., 2011. Google search volume and its influence on liquidity and returns of German stocks. Financ. Mark. Portf. Manag. 25, 239.
Barber, B.M., Odean, T., 2008. All that glitters: the effect of attention and news on the buying behavior of individual and institutional investors. Rev. Financ. Stud. 21, 785–818.
Bleher, J., Dimpfl, T., 2019. Today I got a million, tomorrow, I don't know: on the predictability of cryptocurrencies by means of Google search volume. Int. Rev. Financ. Anal. 63, 147–159.
Bleher, J., Dimpfl, T., 2022. Knitting multi-annual high-frequency google trends to predict inflation and consumption. Econ. Stat. 24, 1–26.
Carhart, M., 1997. On persistence in mutual fund performance. J. Financ. 52, 57–82.
Chai, D., Dai, M., Gharghori, P., Hong, B., 2021. Internet search intensity and its relation with trading activity and stock returns. Int. Rev. Financ. 21, 282–311.
Cheah, E.-T., Fry, J., 2015. Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin. Econ. Lett. 130, 32–36.
Corbet, S., Eraslan, V., Lucey, B., Sensoy, A., 2019. The effectiveness of technical trading rules in cryptocurrency markets. Financ. Res. Lett. 31, 32–37.
Da, Z., Engelberg, J., Gao, P., 2011. In search of attention. J. Financ. 66, 1461–1499.
Dastgir, S., Demir, E., Downing, G., Gozgor, G., Lau, C.K.M., 2019. The causal relationship between Bitcoin attention and Bitcoin returns: evidence from the Copula-based Granger causality test. Financ. Res. Lett. 28, 160–164.
Fama, E.F., 1970. Efficient capital markets: a review of theory and empirical work. J. Financ. 25, 383–417.
Fama, E.F., French, K.R., 1992. The cross-section of expected stock returns. J. Financ. 47, 427–465.
Fama, E.F., Macbeth, J.D., 1973. Risk, return, and equilibrium: empirical tests. J. Political Econ. 81, 607–636.
Gerritsen, D.F., Bouri, E., Ramezanifar, E., Roubaud, D., 2020. The profitability of technical trading rules in the Bitcoin market. Financ. Res. Lett. 34, 101263.

Goyal, A., Jegadeesh, N., 2018. Cross-sectional and time-series tests of return predictability: what is the difference? Rev. Financ. Stud. 31, 1784–1824.
Grobys, K., Ahmed, S., Sapkota, N., 2020. Technical trading rules in the cryptocurrency market. Financ. Res. Lett. 32, 101396.
Grossman, S.J., Stiglitz, J.E., 1980. On the impossibility of informationally efficient markets. Am. Econ. Rev. 70, 393–408.
Härdle, W.K., Harvey, C.R., Reule, R.C., 2020. Understanding cryptocurrencies. Oxford University Press.
Harvey, C.R., Siddique, A., 2000. Conditional skewness in asset pricing tests. J. Financ. 55, 1263–1295.
Hoang, L.T., Baur, D.G., 2022. Loaded for bear: bitcoin private wallets, exchange reserves and prices. J. Bank. Financ. 144, 106622.
Huberman, G., Regev, T., 2001. Contagious speculation and a cure for cancer: a nonevent that made stock prices soar. J. Financ. 56, 387–396.
Ioannidis, J.P., 2022. The end of the COVID-19 pandemic. Eur. J. Clin. Investig. 52, e13782.
Joseph, K., Wintoki, M.B., Zhang, Z., 2011. Forecasting abnormal stock returns and trading volume using investor sentiment: evidence from online search. Int. J. Forecast. 27, 1116–1127.
Kahneman, D., 1973. Attention and effort. Prentice-Hall, Englewood Cliffs, NJ.
Kristoufek, L., 2013. BitCoin meets Google Trends and Wikipedia: Quantifying the relationship between phenomena of the Internet era. Sci. Rep. 3, 3415.
Li, Y., Urquhart, A., Wang, P., Zhang, W., 2021. MAX momentum in cryptocurrency markets. Int. Rev. Financ. Anal. 77, 101829.
Li, Y., Zhang, W., Xiong, X., Wang, P., 2020. Does size matter in the cryptocurrency market? Appl. Econ. Lett. 27, 1141–1149.
Lin, Z.-Y., 2021. Investor attention and cryptocurrency performance. Financ. Res. Lett. 40, 101702.
Liu, Y., Tsyvinski, A., Wu, X., 2022. Common risk factors in cryptocurrency. J. Financ. 77, 1133–1177.
Liu, Y., Tsyvinski, A., 2021. Risks and returns of cryptocurrency. Rev. Financ. Stud. 34, 2689–2727.
Lo, A.W., 2004. The adaptive market hypothesis: market efficiency from evolutionary perspective. J. Portf. Manag. 30, 15–29.
López-Martín, C., Benito Muela, S., Arguedas, R., 2021. Efficiency in cryptocurrency markets: new evidence. Eurasia Econ. Rev. 11, 403–431.
Naeem, M.A., Mbarki, I., Shahzad, S.J.H., 2021. Predictive role of online investor sentiment for cryptocurrency market: evidence from happiness and fears. Int. Rev. Econ. Financ. 73, 496–514.
Nasir, M.A., Huynh, T.L.D., Nguyen, S.P., Duong, D., 2019. Forecasting cryptocurrency returns and volume using search engines. Financ. Innov. 5, 1–13.
Newey, W.K., West, K.D., 1987. Hypothesis testing with efficient method of moments estimation. Int. Econ. Rev. 777–787.
Nguyen, C., Hoang, L., Shim, J., Truong, P., 2020. Internet search intensity, liquidity and returns in emerging markets. Res. Int. Bus. Financ. 52, 101166.
Noda, A., 2021. On the evolution of cryptocurrency market efficiency. Appl. Econ. Lett. 28, 433–439.
Ozdamar, M., Akdeniz, L., Sensoy, A., 2021. Lottery-like preferences and the MAX effect in the cryptocurrency market. Financ. Innov. 7, 1–27.
Panagiotidis, T., Stengos, T., Vravosinos, O., 2019. The effects of markets, uncertainty and search intensity on bitcoin returns. Int. Rev. Financ. Anal. 63, 220–242.
Pástor, Ľ., Vorsatz, M.B., 2020. Mutual fund performance and flows during the COVID-19 crisis. Rev. Asset Pricing Stud. 10, 791–833.
Polyzos, E., Rubbaniy, G., Mazur, M., 2024. Efficient market hypothesis on the blockchain: a social-media-based index for cryptocurrency efficiency. Financ. Rev.
Shen, D., Urquhart, A., Wang, P., 2019. Does twitter predict Bitcoin? Econ. Lett. 174, 118–122.
Smales, L.A., 2022. Investor attention in cryptocurrency markets. Int. Rev. Financ. Anal. 79, 101972.
Svogun, D., Bazán-Palomino, W., 2022. Technical analysis in cryptocurrency markets: do transaction costs and bubbles matter? J. Int. Financ. Mark., Inst. Money 79, 101601.
Takeda, F., Wakao, T., 2014. Google search intensity and its relationship with returns and trading volume of Japanese stocks. Pac. -Basin Financ. J. 27, 1–18.
Urquhart, A., 2016. The inefficiency of Bitcoin. Econ. Lett. 148, 80–82.
Wei, W.C., 2018. Liquidity and market efficiency in cryptocurrencies. Econ. Lett. 168, 21–24.
Yermack, D., 2015. Is Bitcoin a real currency? An economic appraisal. Handbook of digital currency. Elsevier.
Youssef, M., Waked, S.S., 2022. Herding behavior in the cryptocurrency market during COVID-19 pandemic: the role of media coverage. North Am. J. Econ. Financ. 62, 101752.
Yung, K., Nafar, N., 2017. Investor attention and the expected returns of reits. Int. Rev. Econ. Financ. 48, 423–439.
Zhang, W., Li, Y., Xiong, X., Wang, P., 2021. Downside risk and the cross-section of cryptocurrency returns. J. Bank. Financ. 133, 106246.
Zhang, W., Li, Y., 2023. Liquidity risk and expected cryptocurrency returns. Int. J. Financ. Econ. 28, 472–492.